

From Bids to Prices: Understanding the Day-Ahead Market and Simulating Its Clearing

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What is the Day-Ahead Market (DAM)?

Every day at 12:00 CET, electricity prices and traded volumes are set across Europe for the following day.

25

countries

1840

TWh in 2024

5

TWh/day

2m 15s

average run time

15min

Market Time Unit

96

periods/day



Contents

Today's objective is to understand how to simulate the day-ahead market.

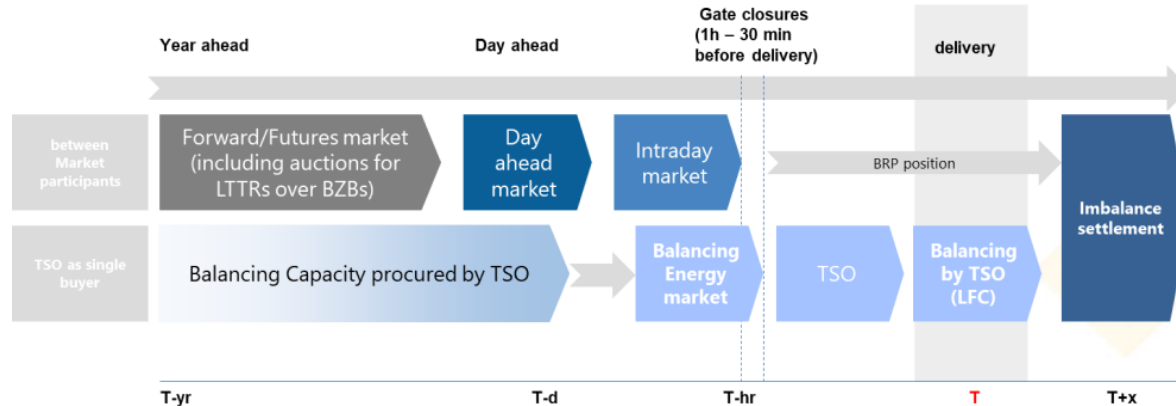
1. Schedule
2. How it works
3. Bidding zones
4. Bid types
5. Market clearing
6. Simulator demo



1. Schedule

DAM is linked to other markets

- Before: future & bilateral contracts
- After: intraday markets & balancing services

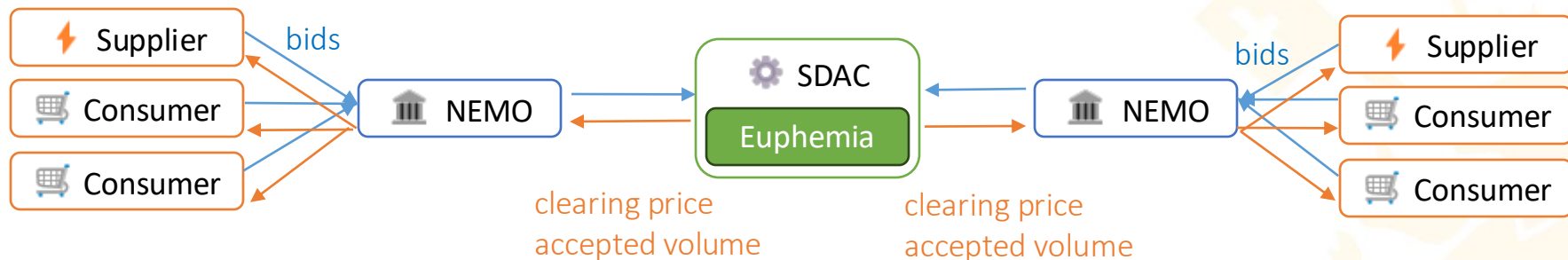


2. How it works

  Market participants (electricity suppliers and consumers) send their bids to the NEMO.

 Nominated Electricity Market Operators (**NEMO**) collect participants' bids and submit them to the SDAC

-  Single Day-Ahead Coupling (**SDAC**) mechanism
 - o collects all bids from all the NEMOs,
 - o runs **marginal pricing** clearing algorithm, **Euphemia**, and
 - o returns clearing prices and accepted volumes



2. Marginal pricing scheme

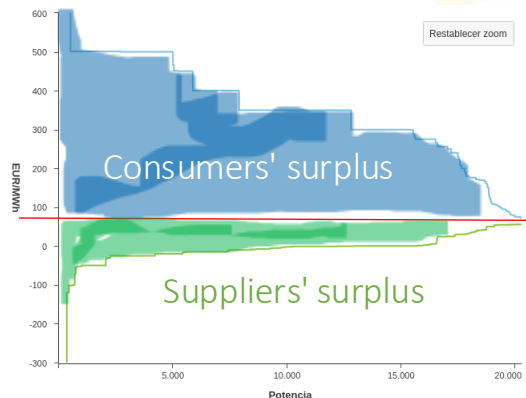
Euphemia is the algorithm used to clear the day-ahead market and follows a marginal pricing scheme.

For each 96 x 15m period:

1. Buy and sell bids are sorted by economic merit
2. The **last accepted bid** sets the clearing price
3. Sell (buy) bids below (above) the clearing price are accepted

All participants receive or pay this **same clearing price**

The difference between the clearing price and the bid price is the **economic surplus**



3. Bidding zones



- The day-ahead market is divided into **bidding zones**.
- The cross-border capacity is limited.
 - If this limit is not reached, different zones are treated as a single market.
 - When cross-border connections become **saturated**, zones are cleared separately and different market prices emerge.

Bidding zones



4. Simple bids

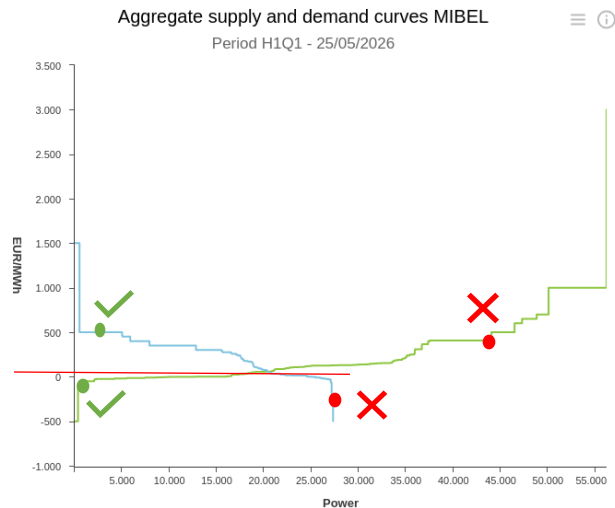
A participant can submit a simple bid consisting of

- ⚡ a power quantity
- 💰 at a price limit
- 🕒 for a specific period of the day

A simple sell (buy) bid is accepted if its price limit is below (above) the clearing price

These bids are useful for flexible technologies that can rapidly adjust their output, such as solar power plants.

Other technologies, such as nuclear plants or combined-cycle units, cannot rapidly change their power output between consecutive bidding periods.



How can we ensure a feasible generation schedule for each unit?

4. Some technological constraints



Technical minimum

the lowest stable loading level below which they cannot operate without shutting down



Ramping constraints

how much the output of a generation unit can change between two periods



Fixed costs

business expenses that remain constant regardless of production volume or sales

4. Block orders

A block order is a bid consisting of

1. a power profile over several periods,
2. a price limit and
3. a Minimum Acceptance Ratio (MAR)

Clearing:



Technical minimum

- A sell (buy) block order is accepted if the if the average clearing price is greater (less) than or equal to its limit price.
- A block order can be accepted fully, partially or rejected
- If a block is partially accepted, all hours are cleared with the same 'acceptance ratio', always above the MAR



Ramping constraints

Market	T	T+1	T+2
Clearing Price (€/MWh)	90	30	90

Mean 70

Simple Bids	T	T+1	T+2
Bidded energy (MWh)	100	100	100
Bidded Price (€/MWh)	60	60	60
Cleared energy	Y	N	Y

Block Order	T	T+1	T+2
Bidded energy (MWh)	100	100	100
Bidded Price (€/MWh)	60		
Cleared energy	Y	Y	Y

4. Scalable Complex Orders (SCO)

A group of simple bids (power + price) with associated constraints:

1. Load Gradient constraint,
2. Minimum Acceptance Power (MAP), and/or
3. Minimum Income Conditions (MIC)



Ramping constraints



Technical minimum



Fixed costs

Clearing:

- A sell SCO is accepted if the total income is greater than or equal to
 - the income required by the simple bids (variable term) +
 - the MIC (fixed term)
- Some simple bids inside of the SCO can be rejected as long as the MAP and the Load Gradient is respected

SCO	T	T+1	T+2
Bidded energy (MWh)	100	100	100
Bidded price (€/MWh)	50	50	50
MIC (€)	7,000		
Cleared energy	Y	N	Y

Three periods

Income: $100 \cdot 90 + 0 \cdot 30 + 100 \cdot 90 = 18000$

Cost: $100 \cdot 50 + 0 \cdot 50 + 100 \cdot 50 + 7000 = 17000 \text{ €}$

$18000 > 17000$

Market	T	T+1	T+2
Clearing Price (€/MWh)	90	30	90

4. Scalable Complex Orders (SCO)

What if a Minimum Acceptance Power (MAP) is included?

Market	T	T+1	T+2
Clearing Price (€/MWh)	90	30	90

SCO	T	T+1	T+2
Bidded energy (MWh)	100	100	100
Bidded price (€/MWh)	50	50	50
MAP (MWh)	20	10	20
MIC (€)	7,000		
Cleared energy	Y	Y	Y

Three periods

Income: $100 \cdot 90 + 10 \cdot 30 + 100 \cdot 90 = 18300$

Cost: $100 \cdot 50 + 10 \cdot 50 + 100 \cdot 50 + 7000 = 17500 \text{ €}$

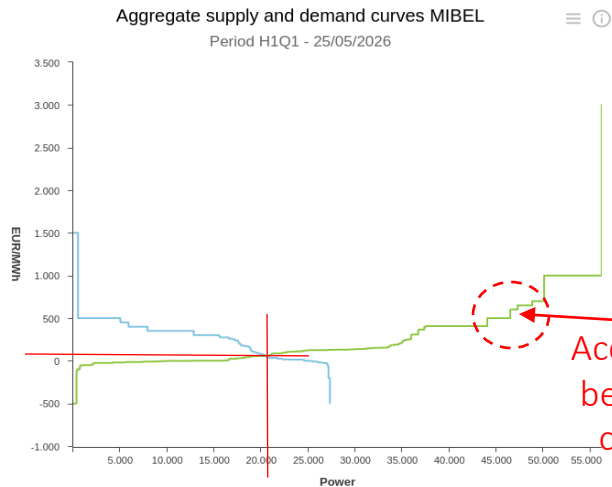
$18300 > 17500$

The acceptance is not continuous ->
MILP

5. Market clearing

How can we determine the last accepted bid when a block order may force some periods to be paradoxically accepted?

The market clearing is an **iterative process** between price determination and the acceptance or rejection of complex orders.



Market	T	T+1	T+2
Clearing Price (€/MWh)	90	30	90

Block Order	T	T+1	T+2
Bidded energy (MWh)	100	100	100
Bidded Price (€/MWh)		60	
Cleared energy	Y	Y	Y

5. Market clearing

Economic Surplus
Maximization
Problem
(Master Problem)

Integer solution

Price
Determination
Sub-Problem

Infeasible solution:

- introduce a cut / prune the node
- back to Master Problem

Feasible integer
solution

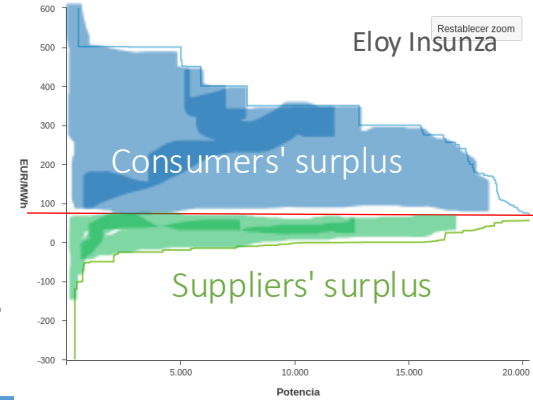
Euphemia aims to maximize economic surplus:
consumers' utility – suppliers' costs

Market	T	T+1	T+2
Clearing Price (€/MWh)	90	30	90

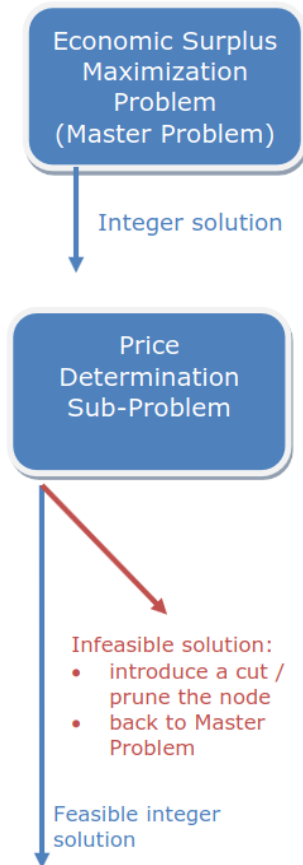
Block Order	T	T+1	T+2
Bidded energy (MWh)	100	100	100
Bidded Price (€/MWh)		60	
Cleared energy	Y	Y	Y

This period contributes with negative surplus, but the whole block has positive surplus

The main challenge is enforcing block orders and SCOs clearing constraints since they depend on the clearing price, an outcome of the problem.

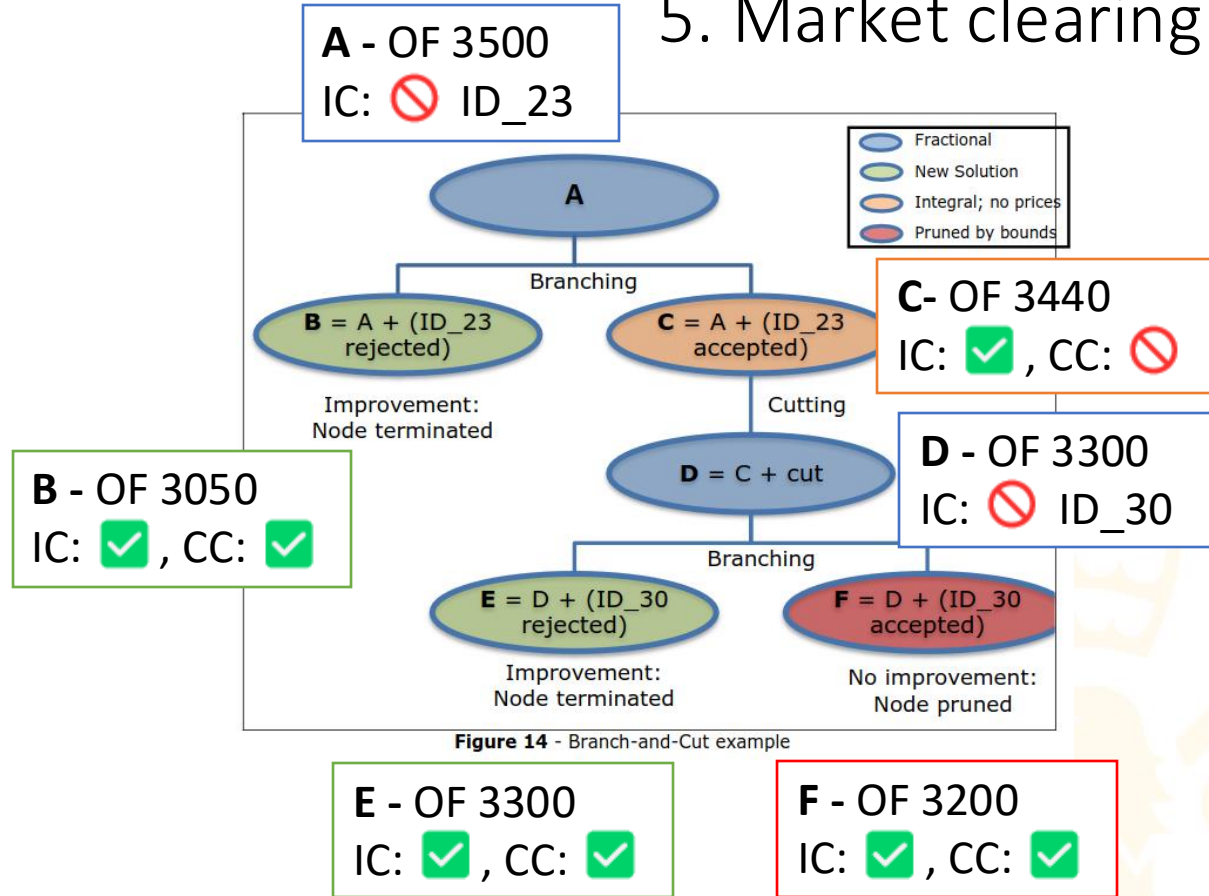
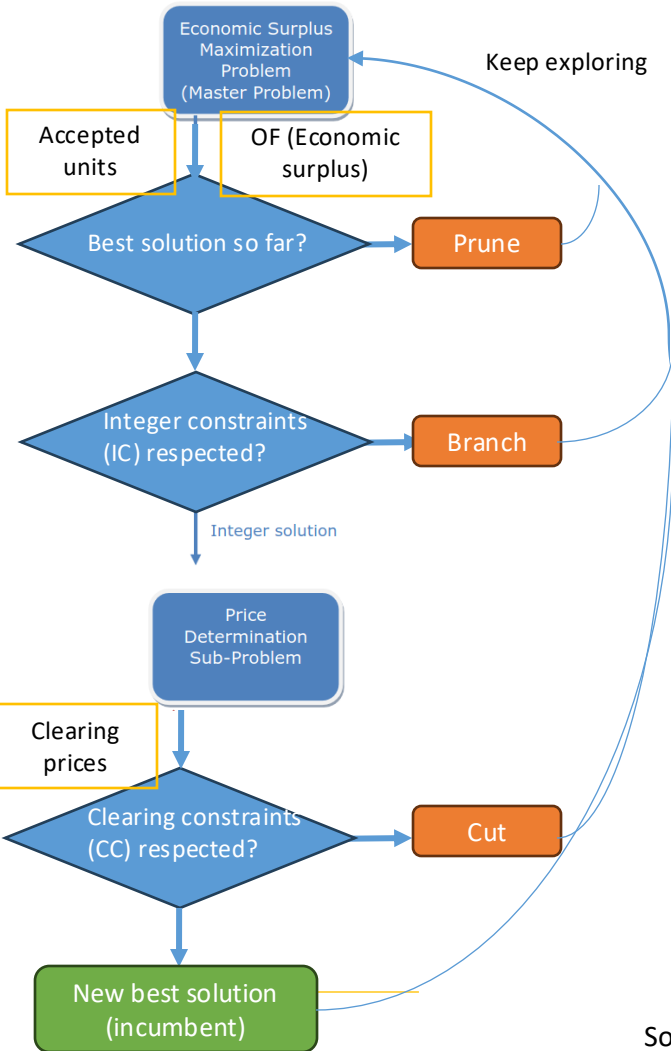


5. Market clearing



- The need of solving the price determination sub-problem to ensure clearing constraints creates an **iterative process**
- The activation of Block Orders and SCO's is controlled by **integer variables**
 - The more relaxed integer constraints, the higher the economic surplus
 - If all integer constraints are relaxed, we get the upper bound of the economic surplus
- We cannot check all integer constraints combinations
 - **Paradoxically accepted orders (PAO)**. Accepted although the economic constraints are not fulfilled. Forbidden.
 - **Paradoxically rejected orders (PRO)**. Refused although the economic constraints are fulfilled. Not desirable.

5. Market clearing



6. Simulator demo

The need:

An open-source simulator to clear the day-ahead market

6. Simulator demo: Market clearing

For n iterations

1. Solve the problem with integer constraints
2. Fix integer constraints and solve the problem again
3. Calculate the **clearing price** using dual variables
4. Check for paradoxically accepted/rejected orders
 - Rank PAOs higher than PROs
 - Rank higher PAOs with higher economic losses
 - Rank higher PROs that would be earning more
5. Force the integer variable of these PAO/PRO to switch and $\rightarrow 1$

6. iberian-day-ahead-market-simulator

- ✓ Computes clearing prices, accepted orders, and cleared energy
- ✓ Handles non-convex/complex orders (Block Orders + SCOs)
- ✓ Works directly with OMIE file formats
- ✓ Compatible with HiGHS and Gurobi



pip install iberian-day-ahead-market-simulator



Code: <https://github.com/EloyID/iberian-day-ahead-market-simulator>



Paper (SSRN, publication in process): <https://dx.doi.org/10.2139/ssrn.6722443>



Validated on 190+ day-ahead markets (Mar 19–Sep 30, 2025) with close agreement vs observed outcomes (prices, Spain–Portugal exchanges, and cleared energy).

6. iberian-day-ahead-market-simulator

1. Load data
2. Change bids or transmission constraints
3. Run the simulation
4. Plot the results



Conclusion

Now we know

- What the DAM is and why it is important
- Its participants
- How a participant can submit its bids
- How the accepted volumes and clearing prices are set
- How to simulate it for your research

Bibliography

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Thank you!

